ASSIGNMENT 2 – CO2

Design and Appraise Black-Box and White-Box Test Cases for a Personal Loan Eligibility Module (LoanEase)

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Date: 05-11-2026 **Total Marks: 50**

1. Equivalence Class Partitioning

ECP divides inputs into groups expected to behave similarly. Age: eligible 21–60; accepted but ineligible 0–20 and 61–120; outside field range below 0 or above 120. Income: eligible at least ₹25,000; ineligible below ₹25,000. Credit score: 600–900 passes the credit rule; 300–599 is rejected only when income is below ₹75,000; score outside 300–900 is outside the stated domain. Amount: ₹50,000–₹5,00,000 and no more than 20 times income; below/above the field range or above 20×income is invalid/ineligible. Examples: age 18 → Age out of range; income ₹20,000 with valid age → Low income; score 550 with income ₹30,000 → Credit risk; amount ₹7,00,000 with income ₹30,000 → Amount too high. Age illustrates the difference between field validation (0–120) and eligibility (21–60).

2. Boundary Value Analysis

BVA tests values at and near limits. Age eligibility: lower 20,21,22 and upper 59,60,61; field limits: -1,0,1 and 119,120,121. Income threshold: ₹24,999, ₹25,000, ₹25,001. Amount limits: ₹49,999, ₹50,000, ₹50,001 and ₹4,99,999, ₹5,00,000, ₹5,00,001; also test 20×income−1, 20×income, 20×income+1. Tenure: 5,6,7; 59,60,61; 83,84,85. For 2-value BVA use boundary and adjacent value (e.g. 24,999/25,000); for 3-value use below, at, above (24,999/25,000/25,001). Counting 10 selected boundary points gives 20 two-value checks or 30 three-value checks; checks may be combined into full test cases with other inputs fixed. BVA detects off-by-one and < versus ≤ mistakes that ECP representatives may miss.

3. Decision Table Testing

Conditions: C1 age 21–60; C2 income ≥₹25,000; C3 score ≥600 OR income ≥₹75,000; C4 amount ≤20×income; C5 tenure ≤60. Reduced rules: R1 C1=No, rest – → reject Age out of range. R2 C1=Yes,C2=No, rest – → reject Low income. R3 C1,C2=Yes,C3=No, rest – → reject Credit risk. R4 C1–C3=Yes,C4=No,C5=– → reject Amount too high. R5 C1–C4=Yes,C5=No → reject Tenure too long. R6 all Yes → Approved. Sample tests respectively: (20,30000,700,100000,36); (30,20000,700,100000,36); (30,30000,550,100000,36); (30,30000,700,700000,36);

(30,30000,700,100000,72); (30,30000,700,100000,36). The approved example has 11.5%; use score 750 to check 9.5%.

4. State Transition Testing

A reasonable model is Draft→Submitted→Under Review. Under Review can move to Approved, Rejected or Cancelled; Approved→Disbursed. Draft and Submitted may also be cancelled. Valid transition tests: Draft/Submit/Submitted; Draft/Cancel/Cancelled; Submitted/Start review/Under Review; Submitted/Cancel/Cancelled; Under Review/Approve/Approved; Under Review/Reject/Rejected; Under Review/Cancel/Cancelled; Approved/Disburse/Disbursed. Invalid tests: Disbursed→Draft, Rejected→Disbursed, Cancelled→Submitted. For invalid transitions, prevent the action, preserve the current state and show a clear message. This is a proposed model; confirm cancellation rules with the actual workflow specification.

5. Random Testing

Generate 1,000 reproducible pseudo-random records using a fixed seed. Use age 0–120, score 300–900, amount ₹50,000–₹5,00,000, tenure 6–84 and a documented income test range (the sheet does not specify its input domain). The oracle independently implements the given rules and compares status, rejection reason and rate. Properties: (1) age below 21 or above 60 always gives Age out of range; (2) with valid age, income below ₹25,000 always gives Low income. Random testing explores combinations but may miss boundaries and rare cases, so combine it with planned ECP, BVA and structural tests.

6. Pairwise Testing

Exhaustive combinations = 3×3×3×3 = 81. A 9-case pairwise set is: (Low,Short,Fixed,Auto-debit); (Low,Medium,Floating,UPI); (Low,Long,Concessional,Cheque); (Mid,Short,Floating,Cheque); (Mid,Medium,Concessional,Auto-debit); (Mid,Long,Fixed,UPI); (High,Short,Concessional,UPI); (High,Medium,Fixed,Cheque); (High,Long,Floating,Auto-debit). Each row gives Amount, Tenure, Interest Type, Repayment Mode. The set covers every pair of values across each pair of parameters. It is smaller because it targets two-way interactions rather than every four-way combination. Higher-order interaction defects may still be missed.

7. Statement and Branch Coverage

There are six decisions (age, income, credit rule, amount, tenure and rate selection), hence 12 decision outcomes. T1 covers false outcomes for all eligibility checks and the lower-rate branch; T2 covers age=true; T3 covers age=false and income=true. Covered branches = 7/12 = 58.3%. Counting 14 executable statements (five eligibility decisions, five rejection returns, rate decision, two rate assignments, final return), 10 are executed = 71.4%. Add four tests: (30,30000,550,100000,36) → Credit

risk; (30,30000,700,700000,36) → Amount too high; (30,30000,700,100000,72) → Tenure too long; (30,30000,750,100000,36) → Approved at 9.5%. These exercise the missing decision outcomes.

8. Condition, Decision and MC/DC

Let A = score<600 and B = income<₹75,000. Tests: T1 score 550/income ₹30,000 gives A=T,B=T, decision=T; T2 score 700/income ₹30,000 gives F,T,F; T3 score 550/income ₹80,000 gives T,F,F. Decision coverage needs two tests (one true and one false). Condition coverage needs three to make both A and B true and false. MC/DC also needs three: T1/T2 proves A independently changes the result while B stays true; T1/T3 proves B independently changes it while A stays true. Four tests would be needed for all combinations; MC/DC is stronger than decision coverage because it proves each condition's independent effect.

9. Control Flow Graph and Cyclomatic Complexity

Decisions are age, income, credit-risk, amount, tenure and rate selection. Expand line 2 into age<21 and age>60, and line 4 into score<600 and income<₹75,000. Predicate count = 8, so for one connected component V(G)=predicate count+1=9. This agrees with E−N+2P when the CFG is drawn with the expanded short-circuit predicates and P=1. If compound conditions are not expanded, only six decision nodes are counted and V appears as 7. Expansion exposes additional short-circuit routes, explaining the difference.

10. Path Testing

Using the expanded CFG, target nine independent paths. Representative tests (age,income,score,amount,tenure): P1 (20,30000,700,100000,36) → age rejection; P2 (61,30000,700,100000,36) → age rejection; P3 (30,20000,700,100000,36) → low income; P4 (30,30000,550,100000,36) → credit risk; P5 (30,30000,700,700000,36) → amount too high; P6 (30,80000,550,100000,72) → tenure too long; P7 (30,30000,700,100000,36) → approved 11.5%; P8 (30,30000,750,100000,36) → approved 9.5%; P9 (30,80000,550,100000,36) → approved 11.5%. A basis path adds at least one new edge. Draw the CFG using the same expanded-predicate convention so its path count matches 9.

11. Test Adequacy and Coverage Gap

100% statement coverage only proves that statements executed; it does not prove assertions would detect incorrect results. To kill the mutant changing income<₹75,000 to income≤₹75,000, use score 550 and income exactly ₹75,000: original condition is false, mutant condition is true. Three useful tests: (1) score 550, income ₹75,000, other inputs valid; (2) age 60, all other inputs valid, to check upper age boundary; (3) income ₹30,000, amount exactly ₹6,00,000 and tenure 60, to check equality at amount and

tenure limits. Require traceability, 100% branch coverage for eligibility decisions, MC/DC for the compound credit rule, boundary tests, and mutation analysis with every surviving mutant reviewed.

12. Integrated Test Suite Design Challenge

A. Build one version-controlled suite and trace requirements to tests: age→ECP/BVA/branch tests; income→ECP/BVA/decision-table tests; credit→decision table/MC/DC; amount and tenure→BVA/decision table; workflow→state-transition tests; EMI parameters→pairwise tests; code logic→statement, branch and path tests. Record requirement ID, test ID, technique, expected result and execution status. B. Report ECP class coverage, BVA boundaries, six decision rules, valid/invalid transitions, all parameter pairs, statement/branch coverage, MC/DC and path coverage. The assignment gives no run results, so actual achieved percentages must be measured. C. Add timeout tests (integration/negative path: safe error, no approval without a score); out-of-range score tests (boundary/robustness: reject invalid score); malformed response tests (API contract: validate fields and types, fail safely). D. For two testers, automate regression and pairwise checks, while focusing manual effort on BVA, loan decision rules and external-service failures. These techniques balance defect detection and effort. Track escaped defects, surviving mutants and release test results each two-week cycle.